

# Revised Generation Interconnection System Impact Study Report

## For Queue Project Q22-2233

Arcogen 138 kV  
February 01, 2025

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2233. The Interconnection Customer proposes a 57.2 MW solar photovoltaic facility interconnecting at the Arcogen 138 kV substation in the Virginia area. The request was received on November 20, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 415 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

This project does not cause any thermal violations under system intact or single contingency conditions. Estimated network upgrade cost: \$0

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

No overloads identified.

#### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

#### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

### **4. Required Network Upgrades and Cost Estimates**

None. Total network upgrade cost allocated to this project: \$0.

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.